

DATABASE MANAGEMENT SYSTEMS (IT 2040)

Question 1

What are database utilities primarily used for?

- A) Creating database schemas
- B) Performing maintenance tasks on created tables and stored data
- C) Writing SQL queries
- D) Designing ER diagrams

Answer: B

Question 2

Which of the following is NOT a data transferring option in SQL Server?

- A) SQL Server Integration Services (SSIS)
- B) BCP (Bulk Copy Program)
- C) BULK INSERT
- D) CREATE TABLE command

Answer: D

Question 3

What does SSIS stand for?

- A) SQL Server Import Services
- B) SQL Server Integration Services
- C) SQL Server Installation System
- D) SQL Server Information Services

Answer: B

Question 4

Which SQL Server edition does NOT include the Import and Export Wizard?

- A) Enterprise Edition
- B) Standard Edition
- C) Express Edition and Local Database Edition
- D) Developer Edition

Answer: C

Question 5

How many main steps are shown in the lecture for exporting data from an Excel file to SQL Server?

- A) 3
- B) 4
- C) 5
- D) 6

Answer: C

Question 6

In the Import/Export Wizard, where do you access the export functionality?

- A) Database menu → Export
- B) Right-click database → Tasks → Export Data
- C) File menu → Export
- D) Tools → Data Export

Answer: B

Question 7

When exporting from Excel, what should you check if column names are in the first row?

- A) Header Row checkbox
- B) First Row Contains Column Names checkbox
- C) Include Headers option
- D) Skip First Row option

Answer: B

Question 8

What does BCP stand for?

- A) Bulk Copy Process
- B) Bulk Copy Program
- C) Batch Copy Protocol
- D) Binary Copy Procedure

Answer: B

Question 9

What is a key advantage of BCP over other data transfer methods?

- A) Graphical user interface
- B) Can transfer large numbers of records efficiently
- C) Requires no configuration
- D) Only works with Excel files

Answer: B

Question 10

Where is BCP installed?

- A) Must be downloaded separately
- B) Only in Enterprise Edition
- C) By default with SQL Server
- D) Part of Windows OS

Answer: C

Question 11

What is the general format of the BCP command?

- A) bcp [database] [action] [file]
- B) bcp {table|view|"query"} {out|in} {data_file}
- C) bcp -source -destination -options
- D) bcp [table] [file] [mode]

Answer: B

Question 12

What does the "out" option in BCP do?

- A) Imports data from file to table
- B) Exports data from table/view to data file
- C) Deletes data
- D) Validates data format

Answer: B

Question 13

What is the difference between "out" and "queryout" in BCP?

- A) No difference
- B) out uses tables/views, queryout uses SQL queries
- C) out is faster
- D) queryout requires admin rights

Answer: B

Question 14

In the BCP command `bcp adventureworks.person out C:\person.csv -T -c -S ServerName`, what does -T mean?

- A) Table mode
- B) Text format
- C) Trusted connection (Windows Authentication)
- D) Transaction mode

Answer: C

Question 15

What does the -c argument specify in BCP?

- A) Connection string
- B) Character data type for operations
- C) Compression mode
- D) Column delimiter

Answer: B

Question 16

What does the -S argument specify in BCP?

- A) Source file
- B) SQL query
- C) SQL Server instance name
- D) Security mode

Answer: C

Question 17

What does BULK INSERT do?

- A) Inserts one row at a time
- B) Loads data from a data file into a table
- C) Creates bulk tables
- D) Exports data to files

Answer: B

Question 18

What is a key difference between BULK INSERT and BCP "in" option?

- A) BULK INSERT is faster
- B) BULK INSERT is executed by SQL Server process; BCP is client-side utility
- C) BCP requires more permissions
- D) No significant difference

Answer: B

Question 19

In BULK INSERT, what does FIELDTERMINATOR specify?

- A) End of file marker
- B) Column delimiter/separator
- C) String terminator
- D) Transaction boundary

Answer: B

Question 20

What does ROWTERMINATOR specify in BULK INSERT?

- A) Last row number
- B) Row separator/end-of-line marker
- C) Maximum rows to import
- D) Row validation rule

Answer: B

Question 21

Given: BULK INSERT emp FROM "D:\data.csv" WITH (FIELDTERMINATOR = ',', ROWTERMINATOR = '\n') What file format is this?

- A) Fixed-width
- B) XML
- C) Comma-separated values (CSV)
- D) JSON

Answer: C

Question 22

What does SELECT INTO do?

- A) Selects data without copying
- B) Copies data from one table to another
- C) Creates indexes
- D) Validates data integrity

Answer: B

Question 23

Which statement about SELECT INTO is TRUE?

- A) Can only work within same database
- B) Can work across linked servers or distributed queries
- C) Cannot create new tables

- D) Only copies table structure, not data

Answer: B

Question 24

What is a key difference between SELECT INTO and BULK INSERT?

- A) SELECT INTO cannot read from external files
- B) BULK INSERT is slower
- C) SELECT INTO requires triggers
- D) No difference

Answer: A

Question 25

Why is database backup important?

- A) To improve query performance
- B) To protect against data loss from failures
- C) To reduce storage space
- D) To speed up transactions

Answer: B

Question 26

Which is NOT a scenario where backups help?

- A) User accidentally drops a table
- B) Hardware failure (disk crash)
- C) Natural disaster
- D) Slow query performance

Answer: D

Question 27

What does a Full Backup include?

- A) Only data files
- B) Entire database including part of transaction log
- C) Only transaction logs

- D) Only changed data since last backup

Answer: B

Question 28

What does a Differential Backup capture?

- A) Entire database
- B) Only transaction logs
- C) Only extents changed since last full backup
- D) Only new records

Answer: C

Question 29

What does a Transaction Log Backup capture?

- A) Complete database
- B) Changed data
- C) Transaction log records since last full or log backup
- D) User transactions only

Answer: C

Question 30

Which backup type allows point-in-time recovery?

- A) Full backup only
- B) Differential backup only
- C) Transaction log backup
- D) None of the above

Answer: C

Question 31

What is a SQL Server Job?

- A) A user query
- B) A specified series of operations performed sequentially
- C) A backup file

- D) A database connection

Answer: B

Question 32

Which of the following can jobs NOT do?

- A) Run on schedules
- B) Run in response to alerts
- C) Replace database administrator
- D) Notify users of status

Answer: C

Question 33

Who can create jobs in SQL Server?

- A) Only sysadmin role members
- B) Only database owners
- C) Users in several roles including sysadmin
- D) Any connected user

Answer: C

Question 34

Who can edit a job after it's created?

- A) Any user
- B) Only job owner or sysadmin members
- C) Only sysadmin
- D) No one - jobs are immutable

Answer: B

Question 35

How can jobs be executed?

- A) Only manually
- B) Only on schedules
- C) According to schedules, in response to alerts, or via sp_start_job stored procedure

- D) Only in response to alerts

Answer: C

SECTION B: MULTIPLE ANSWER QUESTIONS (Select ALL that apply) - 10 Questions

Question 36

Which are valid data transfer methods in SQL Server? **(Select all that apply)**

- A) SQL Server Integration Services (SSIS)
- B) Import/Export Wizard
- C) BCP command
- D) BULK INSERT
- E) SELECT INTO
- F) COPY command

Answers: A, B, C, D, E

Question 37

BCP command options include: **(Select all that apply)**

- A) out - export from table/view to file
- B) in - import from file to table
- C) queryout - export query results to file
- D) format - generate format file
- E) delete - remove data
- F) backup - create backup

Answers: A, B, C, D

Question 38

BULK INSERT command parameters include: **(Select all that apply)**

- A) FROM - specifies source file path
- B) FIELDTERMINATOR - column delimiter
- C) ROWTERMINATOR - row delimiter
- D) BATCHSIZE - rows per batch
- E) MAXERRORS - maximum errors allowed

- F) CREATETABLE - auto-create table

Answers: A, B, C, D, E

Question 39

Types of database backups include: **(Select all that apply)**

- A) Full backup
- B) Differential backup
- C) Transaction log backup
- D) Incremental backup
- E) Partial backup
- F) File/filegroup backup

Answers: A, B, C, E, F

Question 40

Scenarios where backups are essential include: **(Select all that apply)**

- A) User error (accidentally dropping table)
- B) Hardware failure (disk crash)
- C) Natural disasters (fire, flood)
- D) Malware/ransomware attack
- E) Database corruption
- F) Improving query performance

Answers: A, B, C, D, E

Question 41

SQL Server Job characteristics include: **(Select all that apply)**

- A) Series of operations performed sequentially
- B) Can run on one or more schedules
- C) Can respond to alerts
- D) Can run on multiple servers
- E) Automatically notifies users of status
- F) Cannot be disabled once created

Answers: A, B, C, D, E

Question 42

Job execution methods include: **(Select all that apply)**

- A) According to one or more schedules
- B) In response to alerts
- C) Manual execution via sp_start_job
- D) Automatically at SQL Server startup
- E) When CPU is idle
- F) Triggered by user login

Answers: A, B, C, D, E

Question 43

Schedule conditions for jobs include: **(Select all that apply)**

- A) When SQL Server Agent starts
- B) When CPU utilization is idle
- C) One time at specific date/time
- D) Recurring schedule (daily, weekly, monthly)
- E) On database backup completion
- F) When disk space is low

Answers: A, B, C, D

Question 44

Core maintenance plan tasks include: **(Select all that apply)**

- A) Backup databases
- B) Verify database integrity
- C) Maintain indexes (rebuild/reorganize)
- D) Remove old data from msdb
- E) Remove old backups
- F) Optimize queries automatically

Answers: A, B, C, D, E

Question 45

Benefits of maintenance plans include: **(Select all that apply)**

- A) Ensure adequate database performance
- B) Ensure database availability
- C) Automate routine maintenance tasks
- D) Reduce need for manual intervention
- E) Prevent all database errors
- F) Document maintenance activities

Answers: A, B, C, D, F

SECTION C: SCENARIO-BASED & ANALYTICAL QUESTIONS - 5 Questions**Question 46**

Scenario: You need to export 5 million rows from a SQL Server table to a CSV file for analysis in Excel.

Which method is MOST appropriate and why?

- A) Import/Export Wizard - easiest GUI interface
- B) BCP command - optimized for large data volumes
- C) SELECT INTO - direct table copy
- D) Manual SELECT and copy/paste

Answer: B

Question 47

Scenario: You need to load a 10GB CSV file into SQL Server daily at 2 AM.

Best approach:

- A) Use BULK INSERT in a scheduled SQL Server job
- B) Manually run Import/Export Wizard daily
- C) Use BCP interactively each day
- D) Write Python script

Answer: A

Question 48

Scenario: Your production database crashed. You have:

- Full backup from Sunday midnight
- Differential backups daily at noon
- Transaction log backups every hour

Failure occurred Thursday at 3:45 PM. What's the recovery strategy?

- A) Restore Sunday full backup only
- B) Restore Sunday full + Thursday differential
- C) Restore Sunday full + Thursday differential + all hourly log backups from noon to 3:45 PM
- D) Restore only Thursday differential

Answer: C

Question 49

Code Analysis - BCP Command:

```
bcps "SELECT EmployeeID, Name, Salary FROM HR.Employee WHERE DeptID = 5"
queryout C:\dept5.txt -c -T -S SQLSERVER01
```

What does this command do?

- A) Imports data from dept5.txt into Employee table
- B) Exports all employees to dept5.txt
- C) Exports employees from Department 5 to dept5.txt using character format and Windows authentication
- D) Creates a backup of Employee table

Answer: C

Question 50

Maintenance Plan Design:

You're designing a maintenance plan for a 500GB production database used 24/7. Requirements:

- Minimize downtime
- Maximum acceptable data loss: 15 minutes
- Must meet compliance (retain backups 30 days)
- Weekly index maintenance

Which backup strategy is appropriate?

- A) Full backup daily at midnight only
- B) Full backup weekly + Differential daily + Transaction log every 15 minutes
- C) Full backup monthly + Differential weekly
- D) Transaction log backups only every 15 minutes

Answer: B